Institute Affiliated to University of Mumbai)

Mid Semester Examination

Branch	Date	Sem.	Roll No. / Exam Seat No.	Subject	Student's Signature	Junior Supervisor's Name and Sign

Question No.	A	B	C	D	E	F	G	H	Total	Total out of (20 /30 / 40)
1										
2										
3										
4										

CGTE MSE2 - INFT
2025 - 26
29/9/2025

Examiners Signature	Student's Sign (After receiving the assessed answer sheet)

Q6) Difference between local & global illumination models and give importance

(1) Free and open source type I hypervisor providing services that allow multiple computer operating systems to execute.

(2) Xen project runs in more privileged CPU state than any other software on machine

(3) Responsibilities of hypervisor include memory management, launching most privileged domains.

(4) Internet hosting services companies use

1) consider A Point P = (1,2,3) is transformations?
consider A point P = (1,2,3) is transformed by
1) scale by Sx = 2, Sy = 3, Sz = 1
2) Rotate about z-axis 90° anticlockwise
3) translate by (tx = 1, ty = 2, tz = 3) find the coordinate of P'

→

use column vectors in homogeneous coordinate $P = [1,2,3,1]^T$

step 1.

scale S: $S = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Rotate about z by 90° (anticlockwise)

$R_z(90°) = \begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

Translate T by (1,2,3)

$T = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

composite (apply in order) $M = TR_zS$

composite matrix

$M = T \cdot R_z \cdot S$

$$M = \begin{bmatrix} 0 & -3 & 0 & 1 \\ 2 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

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step2 Apply transforms step-by-step

start $p = [1, 2, 3, 1]^T$

step1:

scale:

$$S = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Multiply S.P

$$\begin{bmatrix} 2 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 2\times1 + 0\times2 + 0\times3 + 0\times1 \\ 0\times1 + 3\times2 + 0\times3 + 0\times1 \\ 0\times1 + 0\times2 + 1\times3 + 0\times1 \\ 0\times1 + 0\times2 + 0\times3 + 1\times1 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 3 \\ 1 \end{bmatrix}$$

so after scaling: ps (2, 6, 3)

step2: Rotate about z-axis by 90° anticlockwise

Rotate $R_z \times ps$

$$\begin{bmatrix} 0 & -1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 6 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} 0\times2 + (-1)\times6 + 0\times3 + 0\times1 \\ 1\times2 + 0\times6 + 0\times3 + 0\times1 \\ 0\times2 + 0\times6 + 1\times3 + 0\times1 \\ 0\times2 + 0\times6 + 0\times3 + 1\times1 \end{bmatrix}$$

$$= \begin{bmatrix} -6 \\ 2 \\ 3 \\ 1 \end{bmatrix}$$

After rotation $p_t = [-6, 2, 3, 1)$ only x & y change z remain same

Step3:

Translate : T · Pe

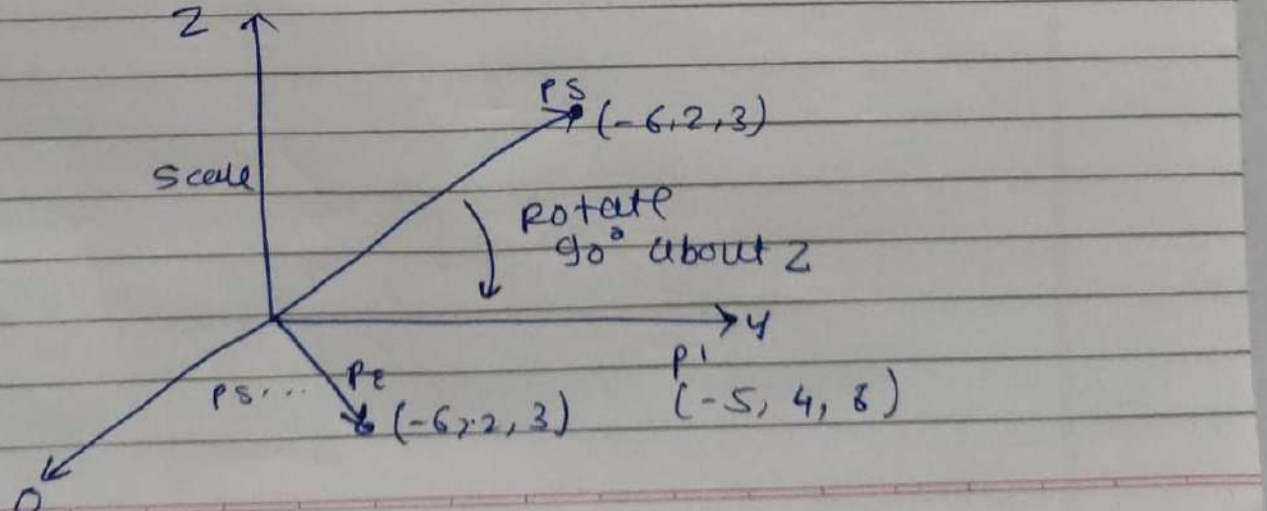
$$\begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -6 \\ 2 \\ 3 \\ 1 \end{bmatrix} = \begin{bmatrix} -5 \\ 4 \\ 6 \\ 1 \end{bmatrix}$$

Final point $P' = (-5, 4, 6)$

Step3 summary

step	operation
0	original
1	scale (2,3,1)
2	Rotate Z 90°
3	translate (1,2,3)

~~(1,2,3)~~ Coordinates
~~(2,6,3)~~ (1, 2, 3)
~~(-6,2,3)~~ (2, 6, 3)
t (-6, 2, 3)
(-5, 4, 6)



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2. Rotation:

Rotation Matrix :-

$$\begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P'' = R_m \times P'$$

$$= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 & 6 & 4 \\ 0.5 & 0.5 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

$$P'' = \begin{bmatrix} -0.5 & -0.5 & -2 \\ 2 & 6 & 4 \\ 1 & 1 & 1 \end{bmatrix}$$

$A'' = (-0.5, 2)$, $B'' = (-0.5, 6)$, $C'' = (-2, 4)$

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3. Translation:

Translation Matrix :-

$$\begin{bmatrix} 1 & 0 & T_x \\ 0 & 1 & T_y \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

$$P''' = T_m \times P''$$

$$= \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -0.5 & -0.5 & -2 \\ 2 & 6 & 4 \\ 1 & 1 & 1 \end{bmatrix}$$

$$P''' = \begin{bmatrix} 3.5 & 3.5 & 2 \\ 0 & 4 & 2 \\ 1 & 1 & 1 \end{bmatrix}$$

$A''' = (3.5, 0)$ $B''' = (3.5, 4)$ $C''' = (2, 2)$

2. Rotation :-

Rotation about Z-axis 90° anti-clockwise:

$$P'' = \begin{bmatrix} x'' \\ y'' \\ z'' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 2 \\ 6 \\ 3 \end{bmatrix} = \begin{bmatrix} -6 \\ 2 \\ 3 \end{bmatrix}$$

$P''(-6, 2, 3)$

3. Translation :-

$t_x = 1$, $t_y = 2$, $t_z = 3$

$$P''' = \begin{bmatrix} x''' \\ y''' \\ z''' \end{bmatrix} = \begin{bmatrix} t_x \\ t_y \\ t_z \end{bmatrix} + \begin{bmatrix} x'' \\ y'' \\ z'' \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} \begin{bmatrix} -6 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} -5 \\ 4 \\ 6 \end{bmatrix}$$

$P'''(-5, 4, 6)$, these are Final coordinates.

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• What is homogeneous coordinate system?

A homogeneous-coordinate system is an extension of a cartesian coordinate system that adds one extra dimension. For 3D space a cartesian point (x, y, z) is represented in homogeneous coordinates as 4D vector (x_h, y_h, z_h, w). In CG, the convention is typically to set $w = 1$, so the point $P(x, y, z)$ is represent as $P(x, y, z, 1)$

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- Final coordinates :

  A (3.5 , 0) , B (3.5 , 4) , C (2, 2)

- Single Composite Matrix = $T_M \times R_M \times S_M$.

$$= \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} * \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \times \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0.5 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -0.5 & 4 \\ 2 & 0 & -2 \\ 0 & 0 & 1 \end{bmatrix}$$

- Why order of matrix multiplication is important?

  As we know, matrix multiplication is not commutative i.e. $A \times B \neq B \times A$. If we perform $A \times B$ is gives another output and $B \times A$ will give another so order of matrix multiplication is important.

Q.2.

Ans.

* Input Matrix = $\begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$

1. Scaling :-

$$P' = \begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = \begin{bmatrix} S_x & 0 & 0 \\ 0 & S_y & 0 \\ 0 & 0 & S_z \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} \cdot \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix} = \begin{bmatrix} 2 \\ 6 \\ 3 \end{bmatrix}$$

P' (2, 6, 3)